

How to Select Your Next Development Board

Selection becomes tough when one has to choose from a list of boards with almost similar capabilities. From designers to young hobbyists, everyone finds it difficult to identify the right board for them. We hope this article will serve as an aid for selecting your next development board

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Built around a microcontroller (MCU) or a microprocessor (MPU), a development board allows a learner to get the correct design, along with the required speed and convenience to create an easy, interesting and foolproof project. The idea behind the creation of a development board was to make the task of a product designer easy. However, over time, this trend has changed. Now development boards are being widely used by hobbyists, young engineers and other learners.

For an amateur or a young engineer who has just started building projects, assembling a bunch of small components on a single hardware in a particular design so as to create a working project is a very tough job. A single mistake in designing the basic board for a project can lead to failure. This is where a development board proves useful. A designer does not have to keep modifying the basic circuit of the project while using a development board. The development board acts as the basic building block, leaving an engineer with time and effort to work on other parts of the project.

Basic uses of a development board

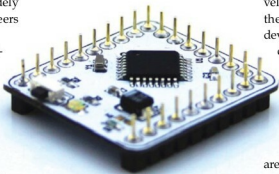
All development boards are used for certain type of work, be it an open source development board or another type that allows modifications in its existing design. Listed below are some uses:

1. To develop the software, bench-

mark and evaluate the MCU/MPU used. The software is developed to enable the MCU/MPU to work with the different on-board components in order to make the board suitable for the basic operations of any application.

2. To develop the software drivers, prototype and validate them for the new system-on-chip (SoC) intellectual property (IP) blocks, like a video engine or a modem.

3. To test the customised logic blocks, or the system IPs in a field-



programmable gate array (FPGA), connected to the core of the MCU/MPU and running at the speed of an application-specific integrated circuit (ASIC).

4. To shorten the development time by providing a ready-to-use platform for firmware development and data collection.

Of course, there are other uses but these depend upon the application a development board is used for.

Open source development board

The development boards licensed

under the open source licences allow engineers to add some value to their designs by incorporating them in the existing design, giving some credibility to the creator of the original board. These boards have circuits that can be modified or improvised and rebuilt. They facilitate the involvement of various open source communities with the associated engineers in order to provide their views on every aspect starting from selecting the board to creating a working model.

Working with an open source development board becomes easier with the availability of free drivers and development tools that can be easily downloaded from Internet.

Some popular boards in this category are the i.MX53, Beagle-Bone Black, Raspberry Pi, Arduino UNO, Gooseberry, Origen, Snowball and Intel Galileo. The list of boards is quite long and additions are made to it every year. These open source development boards provide an open design, size, processing speed and a great price to start with. Other features include the open source software tools that inspire application developers to build third party applications around the projects based on these boards.

Seetharaman Devendran, general manager, MCU, LPRF, sales & marketing, Texas Instruments, says, "The open source development boards like MSP430 are like cupcakes for the student community as these provide the debugging facility at the initial steps. The other development boards that do not have the open source design do not allow any debugging, and this makes it tough to work with those designs."